1.

Consider the following questions:

a. How many elements does the set $\{0, 2, 4, 6\} \cup \{0, 3, 6\}$ have, and what are they?

5 elements, $\{0, 2, 4, 3, 6\}$

b. How many elements does the set $\{0, 2, 4, 6\} - \{0, 3, 6\}$ have, and what are they?

3 elements, {2,4,6}

2.

Over the universe $\mathcal{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, consider the sets $A = \{1, 4, 7, 10\}$, $B = \{1, 2, 3, 4, 5\}$, and $C = \{2, 4, 6, 8\}$. What are the following sets?

a. $\overline{A} = \{2, 3, 5, 6, 8, 9\}$

b. $A \cup \overline{B} = \{1, 4, 6, 7, 8, 9, 10\}$

c. $\overline{B} \cap (C - A) = \{6, 8\}$

d. $\overline{A \cap B} \cup C = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$

3. Let $\mathcal{U}$ denote the universe set and $A, B \subseteq \mathcal{U}$. Show that if $\overline{A} \subseteq \overline{B}$, then $B \subseteq A$.

- Let $x \in \overline{A} \subseteq \overline{B}$, where x is an arbitrary element.
- $\forall x. \left(x \in \overline{A} \Rightarrow x \in \overline{B} \right)$
- $\forall x. (x \in B \Rightarrow x \in A)$
- $x \in B \subseteq A$
- $B \subseteq A$

4.

Here are some properties of sets that do not hold in general. What can we deduce about sets A, B from these properties, if they do hold?

a. "$A \cap B = A$": If this holds, then $A = B$

b. "$A \cup B = A$": If this holds, then $B = \varnothing$

c. "$\overline{A} \cup B = \varnothing$": If this holds, then $B = \varnothing$

d. "$\overline{A \cup B} = \overline{B}$": If this holds, $A \cap \overline{B} = \varnothing$

5.

State De Morgan's law for sets without looking up the definition. Write the proof of De Morgan's law for sets.

De Morgan's Law: $\overline{A \cup B} \equiv \overline{A} \cap \overline{B}$ and $\overline{A \cap B} \equiv \overline{A} \cup \overline{B}$

- Case 1: $\overline{A \cup B} \equiv \overline{A} \cap \overline{B}$
- Let $x \in \overline{A \cup B}$, where x is an arbitrary element.
- $x \notin A \cup B$
- $x \notin A \wedge x \notin B$
- $\neg(x \in A \vee x \in B)$
- $x \in \overline{A} \wedge x \in \overline{B}$
- $x \in \overline{A} \cap \overline{B}$
- $\top$
- Case 2: $\overline{A \cap B} \equiv \overline{A} \cup \overline{B}$
- Let $x \in \overline{A \cap B}$, where x is an arbitrary element.
- $x \notin A \cap B$
- $x \notin A \cap x \notin B$
- $\neg(x \in A \wedge x \in B)$
- $x \in \overline{A} \vee x \in \overline{B}$
- $x \in \overline{A} \cup \overline{B}$
- $\top$

6. Which of the following laws are true for all sets A, B, C? Prove or disprove each one.

a. $A \cap B = B \cap A$

- Let $x \in A \cap B$, where x is an arbitrary element.
- $x \in A \wedge x \in B$
- $x \in B \wedge x \in A$
- $x \in B \cap A$
- $\top$

b. $A - B = B - A$

- Let $A = \{1, 2, 3, 4, 5\}$
- Let $B = \{1, 2, 3, 6\}$
- $A - B = \{4, 5\}$
- $B - A = \{6\}$
- $\bot$

c. $(A - B) \cap C = (A \cap C) - B$

- Case 1:
- Let $x \in (A - B) \cap C$, where x is an arbitrary element.
- $x \in (A \wedge \neg B) \wedge C$
- $x \in A \wedge \neg B \wedge C$
- $x \in (A \wedge C) \wedge \neg B$
- $x \in (A \cap C) - B$
- Case 2:
- Let $x \in (A \cap C) - B$, where x is an arbitrary element.
- $x \in (A \wedge C) \wedge \neg B$
- $x \in A \wedge C \wedge \neg B$
- $x \in A \wedge \neg B \wedge C$
- $x \in (A - B) \cap C$

d. $A - (B \cap C) = (A - B) \cap (A - C)$

- Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- Let $B = \{4, 5, 6, 7\}$
- Let $C = \{1, 2, 3, 4, 5\}$
- $A - (B \cap C) = \{1, 2, 3, 8, 9, 10\}$
- $(A - B) \cap (A - C) = \{8, 9, 10\}$
- $\perp$

7.

The symmetric difference of two sets is defined as: $A \Delta B = (A - B) \cup (B - A)$.

Prove the following properties for all sets A, B, C:

a. $A \Delta B = B \Delta A$

- Let $x \in A \Delta B$, where x is an arbitrary element.
- $x \in (A - B) \cup (B - A)$
- $x \in (A - B) \vee x \in (B - A)$
- $x \in A \wedge \neg B \vee x \in B \wedge \neg A$
- $x \in B \wedge \neg A \vee x \in A \wedge \neg B$
- $x \in (B - A) \vee x \in (A - B)$
- $x \in (B - A) \cup (A - B)$
- $x \in B \Delta A$
- Hence, $A \Delta B = B \Delta A$

b. $(A\Delta B)\Delta C = A\Delta(B\Delta C)$

- Let $x \in (A\Delta B)\Delta C$, where x is an arbitrary element.
- $x \in ((A - B) \cup (B - A))\Delta C$
- $x \in (((A - B) \cup (B - A)) - C) \cup (C - ((A - B) \cup (B - A)))$
- $x \in (((A - B) \cup (B - A)) - C) \vee x \in (C - ((A - B) \cup (B - A)))$
- $x \in (((A \wedge \neg B) \vee (B \wedge \neg A)) \wedge \neg C) \vee x \in (C \wedge \neg((A \wedge \neg B) \vee (B \wedge \neg A)))$
- $x \in (((A \wedge \neg B) \vee (B \wedge \neg A)) \wedge \neg C) \vee x \in (C \wedge ((\neg A \vee B) \wedge (\neg B \vee A)))$
- $x \in (A \wedge \neg B \wedge \neg C) \vee (B \wedge \neg A \wedge \neg C) \vee (C \wedge \neg A \wedge \neg B) \vee (C \wedge A \wedge B)$
- $x \in (A \wedge ((\neg B \wedge \neg C) \vee (B \wedge C))) \vee (\neg A \wedge ((B \wedge \neg C) \vee (\neg B \wedge C)))$
- $x \in (A \wedge \neg(B\Delta C)) \vee (\neg A \wedge (B\Delta C))$
- $x \in (A - (B\Delta C)) \cup ((B\Delta C) - A)$
- $x \in A\Delta(B\Delta C)$
- $\top$

c. $A\Delta A = \varnothing$

- Let $x \in A\Delta A$, where x is an arbitrary element.
- $x \in (A - A) \cup (A - A)$
- $x \in \varnothing \cup \varnothing$
- $x \in \varnothing$
- $\top$